

26STEMSCATBP3

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Problem

Let $\sum_{\ell=0}^{2^{2025}} a_{2\ell+1} x^{2\ell+1}$ be a polynomial of degree $2^{2026} + 1$ with integer coefficients, such that

$$2\mathcal{P}(x) = (x + \sqrt{x^2 - 1})^{2^{2026}+1} + (x - \sqrt{x^2 - 1})^{2^{2026}+1}$$

$\forall x \in \mathbb{R}$.

Prove that $8 \nmid a_1$ and $v_2(a_{2\ell+1}) \geq 2026$ for all integers ℓ with $1 < \ell \leq 2^{2025}$, where $v_2(n)$ is the largest integer k such that 2^k divides n

Solution. Let $T_n(x)$ be the Chebyshev Polynomials of the first kind (i.e. $T_0(x) = 1, T_1(x) = x$ and $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$). As a result, by definition

$$P(x) = T_{2^{2026}+1}(x)$$

Claim 1 — For every integer $n \geq 1$ and every integer j with $0 \leq j \leq \lfloor n/2 \rfloor$,

$$\sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} = \frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1}.$$

Proof. We will proceed with induction, say it works for all positive integers less than n .

The base case is trivial

We begin by rewriting the binomial coefficients as

$$\binom{n}{2k} = \frac{n}{2k} \binom{n-1}{2k-1}, \quad \binom{k}{j} = \frac{k}{j} \binom{k-1}{j-1}.$$

Substituting into the sum gives

$$\sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} = \frac{n}{2j} \sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n-1}{2k-1} \binom{k-1}{j-1}.$$

Now perform the change of variables $m = k - 1$. Then m runs from $j - 1$ to $\lfloor (n - 2)/2 \rfloor$, and the sum becomes

$$\frac{n}{2j} \sum_{m=j-1}^{\lfloor (n-2)/2 \rfloor} \binom{n-1}{2m+1} \binom{m}{j-1}.$$

We now apply induction

$$\sum_{m=r}^{\lfloor N/2 \rfloor} \binom{N}{2m+1} \binom{m}{r} = \binom{N-r}{r} 2^{N-2r-1},$$

valid for all integers $N \geq 1$ and $r \geq 0$. Taking $N = n - 1$ and $r = j - 1$, we obtain

$$\sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} = \frac{n}{2j} \binom{n-j}{j-1} 2^{n-2j}.$$

Finally, using the identity

$$\binom{n-j}{j-1} = \frac{j}{n-j} \binom{n-j}{j},$$

we conclude that

$$\sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} = \frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1},$$

as claimed. □

Claim 2 — For every integer $n \geq 1$,

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1} x^{n-2j},$$

where

$$T_n(x) = \frac{(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n}{2}.$$

Proof. Let

$$A = x + \sqrt{x^2 - 1}, \quad B = x - \sqrt{x^2 - 1}.$$

Then $AB = 1$ and $A + B = 2x$.

Expanding both terms using the binomial theorem,

$$A^n + B^n = \sum_{m=0}^n \binom{n}{m} x^{n-m} (\sqrt{x^2 - 1})^m + \sum_{m=0}^n \binom{n}{m} x^{n-m} (-\sqrt{x^2 - 1})^m.$$

All odd powers of $\sqrt{x^2 - 1}$ cancel, leaving

$$A^n + B^n = 2 \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} x^{n-2k} (x^2 - 1)^k.$$

Dividing by 2, we obtain

$$T_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} x^{n-2k} (x^2 - 1)^k.$$

Next, expand

$$(x^2 - 1)^k = \sum_{j=0}^k \binom{k}{j} (-1)^j x^{2k-2j}.$$

Substituting into (1),

$$T_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} \sum_{j=0}^k \binom{k}{j} (-1)^j x^{n-2j}.$$

By definition of the inner sum, this can be written as a double sum:

$$T_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \sum_{j=0}^k \binom{n}{2k} \binom{k}{j} (-1)^j x^{n-2j}.$$

Let

$$S = \{(k, j) \in \mathbb{Z}^2 : 0 \leq j \leq k \leq \lfloor n/2 \rfloor\}.$$

Then the right-hand side of (2) is precisely

$$\sum_{(k,j) \in S} \binom{n}{2k} \binom{k}{j} (-1)^j x^{n-2j}.$$

For a fixed integer j , the condition $(k, j) \in S$ is equivalent to

$$j \leq k \leq \lfloor n/2 \rfloor.$$

Therefore, the sum in (3) may be regrouped according to the value of j , giving

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} \sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} (-1)^j x^{n-2j}.$$

For fixed j , the factor $(-1)^j x^{n-2j}$ does not depend on k . Hence, by the distributive law,

$$\sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j} (-1)^j x^{n-2j} = (-1)^j x^{n-2j} \sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j}.$$

Substituting this into (4), we obtain

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j x^{n-2j} \sum_{k=j}^{\lfloor n/2 \rfloor} \binom{n}{2k} \binom{k}{j}.$$

This justifies the reordering of the summation.

So we can conclude that:

For every integer $n \geq 1$,

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j \frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1} x^{n-2j}$$

□

For $j < 2^{2025}$,

$$\nu_2 \left(\frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1} \right) = s(n-2j) + s(j) - s(n-j) + n-2j-1 - \nu_2(n-j)$$

where s denotes the sum of digits in base 2. The result is due to Legendre.

Since $\frac{n}{n-j} \binom{n-j}{j}$ is an integer, for $j \leq 2^{2025} - 1013$, $\nu_2(a_{n-2j}) \geq 2026$.

Now say $j = 2^{2025} - i$ for some $0 < i < 1013$.

As a result, we have to prove that

$$s(2i+1) + s(2^{2025} - i) - s(2^{2025} + i + 1) + 2i + 1 - \nu_2(i+1) \geq 2026$$

Say i has a total of k digits with m many 1s and t many ending 0s. Note that $s(i) = m = s(2i)$ and $\nu_2(i) = t$.

Note that $k \leq 10$ since $i \leq 1012$.

So, $s(2^{2025} + i) = 1 + s(i) = 1 + m$. $s(2^{2025} - i) = 2025 - m - t + 1$

$$\begin{aligned} & s(2i+1) + s(2^{2025} - i) - s(2^{2025} + i + 1) + 2i + 1 - \nu_2(i+1) \\ &= m + 1 + 2025 - m - t + 1 - 1 - m + 2i - t \\ &= 2026 + 2i - m - 2t \end{aligned}$$

Due to size reasons, $2i - m - 2t \geq 0$ for large enough i and for small i it can be easily verified.

This concludes the fact that

$$\nu_2 \left(\frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1} \right) \geq 2026$$

For $j = 2^{2025}$,

$$\nu_2 \left(\frac{n}{n-j} \binom{n-j}{j} 2^{n-2j-1} \right) = \nu_2 \left(\binom{2^{2025} + 1}{2^{2025}} \right) + 0 = 0$$

So $8 \nmid a_1$. This concludes our problem.